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Volumetric Analyses of Monotreme Brains

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INTRODUCTION

The volumes of eleven brain structures — (medulla oblongata, cerebellum, mesencephalon, diencephalon, and the following divisions of the telencephalon; neocortex, schizocortex, hippocampus, septum, paleocortex-amygdala complex, bulbus olfactorius and the striatum) of the platypus and of the echidna were calculated. The volumes of the same eleven structures have been calculated previously for insectivores, bats and primates (Stephan and Andy 1970, Stephan and Pirlot 1970, and Pirlot and Pottier 1977). Complete details of the methods used can be found in these publications. In this paper we comment briefly on the shape of some of these structures in the monotremes and especially on some of the histological characteristics of the cortex.

The organization of the cortex in Monotremes has been described in detail by Abbie (1940). We will not comment here on his morphological subdivisions, although we found it somewhat difficult at times to recognise in our specimens all the variations that he has distinguished. Only two cortical layers are of special interest: layer II, which is well marked, as is usual in mammals, and layer V (which may appear to be layer IV in several places, due to the indistinctiveness of III and IV).

MATERIAL AND METHODS

One specimen of *Ornithorhynchus anatinus* (body weight 1040 g, brain weight 9.169 g) and one of *Tachyglossus aculeatus* (body weight 4720 g, brain weight 27.518 g) were perfused with saline and then 20% formalin. They were stored for a week in 10% formalin, then dissected out, weighed, embedded in paraffin wax, and cut serially at 10 μ . Every fifth section was stained with Nissl. For each species, 100-200 of these sections were photographed and the measurements of area and the volumes were made from these photographs (See Stephan and Pirlot, 1970 for details). These photographs were also used in the estimates of the length of some brain structures.

For comparison, the brain of one *Didelphis marsupialis* (body weight 1205 g, brain weight 6.885 g) was treated in the same way.

The measurements of cells were made using a Zeiss-Jena NU 2 microscope with a micrometer in a 12.5 x eyepiece and with a 50 x objective.

HISTOLOGICAL FEATURES OF THE CORTEX

The platypus brain is lissencephalic, the rhinal fissure being the only land mark on its surface. In contrast, the echidna brain is gyrencephalic with well-marked fissures. Neither brain, unlike those of many lower mammals, shows a clear separation into neocortex (isocortex), paleocortex (allocortex), or mesocortex (juxtallocortex). This is of some interest as the *Didelphis* brain does show such subdivisions, particularly a well differentiated intermediate area (mesocortex) described by Oswaldo-Cruz and Rocha-Miranda (1968) as a piriform cortex. Thus, there is less histological differentiation of the monotreme cortex than there is in *Didelphis*.

Platypus

Under the outer plexiform layer I, there is a well marked layer II, which is 60-70 μ thick and in which the rounded medium sized cells (larger than granule cells) are densely packed. The dense layer V has an irregular width, and contains rounded or ovoid cells and a considerable number of larger, usually ovoid neurons. Some of the larger cells have subpyramidal shapes. The number of these cells in a one centimetre square of the micrometer at 312.5 x magnification is around 20 in the platypus (cf. 45 in the echidna).

An interesting feature of the cortex, is the existence of a well developed schizocortex-like area in the vicinity of the hippocampus, i.e. in the subicular-

TABLE 1

A. CELL SIZES IN PLATYPUS (SECTION 662)			
	Mean diameters	Diameter ranges	Mean smaller/greater
Layer II	11.90 x 10.43	16.24 - 6.09	87.6%
Layer V	12.11 x 9.38	22.33 - 6.09	77.4%
B. CELL SIZES IN ECHIDNA (SECTION 1656)			
	Mean diameters	Diameter ranges	Mean smaller/greater
Layer II	12.26 x 8.36	20.30 - 6.09	68.2%
Layer V	14.75 x 10.89	24.36 - 6.09	73.8%
C. CELL SIZES IN DIDELPHIS (SECTION TO PLATE 6 IN OSWALDO-CRUZ AND ROCHA-MIRANDA, 1968)			
	Mean diameters	Diameter ranges	Mean smaller/greater
Layer II	16.95 x 10.52	24.36 - 6.09	62.0%
Layer V*	16.70 x 11.11	28.42 - 6.09	66.5%

* Or equivalent in position. In all cases, N = 100. Measurements in μ m. Magnification for measuring: 50 x 12.5 = 625 x. The repetition of some figures (e.g. 6.09) indicates the limit of precision that could be attained in reading the micrometer scale (1 division = 2.03 μ m).

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retrosplenial area. In contrast to the data in bats, there is no schizocortex in the lateral region of brain between palco- and neocortex.

Echidna

Under the outer plexiform layer I, layer II is 70-90 μ thick and contains medium sized cells which are mostly ovoid though sometimes rounded. This layer is usually much less dense and less well marked than the corresponding layer in the platypus, but in some rhinencephalic regions, it is denser and can be separated into a layer IIa, which is narrower and less dense, and a layer IIb, which is broader and denser. Layer V varies considerably in thickness. It contains medium sized cells intermingled with granule cells and with a number of larger neurons, some of which are pyramidal. The number of the large cells in a one centimetre square of the micrometer at 312.5 x magnification is about 45 (cf. 20 in platypus).

In the echidna, as in the platypus, a schizocortex-like area is found in the subicular-retrosplenial area, near both dorsal and ventral extremities of the hippocampal arch, but no lateral schizocortex is present.

CELL MEASUREMENTS

The mean lengths were measured of the two diameters of a random sample of 100 cells from layer II and from layer V at the level of the anterior commissure and the upper lateral (orbital) region of neocortex. The measurements from the three species are shown in Table 1. Here it can be seen that the cells in layer II are more rounded in the platypus than in the echidna. The smaller diameter of the cells is 87.6% of the length of the cell's larger diameter in the platypus while in the echidna, the same proportion is 68.2%. In layer V, the proportion for the platypus is 77.4% and that of the echidna 73.8%. A possible interpretation of measurements, is that in layer II the more rounded cells of the platypus are of a younger shape than those of the echidna, whereas in layer V all cells are of the mature type. This interpretation is supported by the differences in the density of layer II between the platypus and the echidna. The dense cellular arrangement of layer II in the platypus is probably much closer to the embryonic condition than is the more open arrangement of cells in the echidna. In this sense, the platypus cortex could be termed more primitive than that of the echidna.

Didelphis

It is interesting to compare the monotremes with the reputedly primitive, subplacental mammal, *Didelphis*. A detailed description of the brain of *Didelphis* is outside the scope of this paper, but the following points are of interest.

Under the outer plexiform layer, there is a fairly consistent pattern at most levels from rostral to caudal cortex. Using the section upon which plate 6 in Oswaldo-Cruz and Rocha-Miranda (1968) is based, it can be seen, that in the orbital (dorso-lateral) region, layer II is 80-120 μ thick, is moderately dense and contains large ovoid (sometimes rounded) cells. In the piriform (ventro-lateral)

region, this layer is much denser, and two layers can be distinguished — an external layer with rounded cells and an internal one with elongate or spindle-shaped, radially oriented cells.

Undereath layer II, the organisation varies according to the level and to the region. In a general way, especially in the orbital region that most resembles what we have observed in the monotremes, the area corresponding in location to layer V of the platypus and the echidna, shows a moderately dense assemblage of cells of various diameters, generally having a globulous or ellipsoid shape.

The differing diameters of cells shown in Table 1 were tested statistically to see if the differences between layers and between species were significant (Table 2).

TABLE 2
DIFFERENCES BETWEEN GREATER DIAMETERS

Between mean differences tested	Significant (s) or non-significant (ns) by Mann-Whitney test at 5% level
A II - B II	NS
A II - C II	S
B II - C II	S
A V - B V	S
A V - C V	S
B V - C V	S

SHAPE OF BRAIN AND BRAIN STRUCTURES

A rather vague statement was made by Ziehen (1897, p.34) who writes 'Das Kleinhirn (in *Ornithorhynchus*) wird von den Grosshirnhemisphären noch weniger als bei *Echidna* überlagert'. This suggests that in polar view the hemisphere occupies more of the total brain-length in the echidna than it does in the platypus. However, from our serial sections, we calculate that hemispheres of the platypus occupy 82.0% and that of the echidna 81.5% of the length of the cortex plus cerebellum. (There was little difference in the volume shrinkage of the brains as a result of the embedding process — 63.5% for the platypus and 61.5% for the echidna). Thus, the percentage of the brain length occupied by hemispheres and by cerebellum (17%) is the same for both species. What was not described accurately by Ziehen (1897) is that the cerebellum reaches further forward *under* the hemispheres in the platypus than it does in the echidna, so that the real percentage of total length occupied by the cerebellum should be 34% in the platypus and 23% in the echidna. This corresponds well with our volume estimates (Table 3).

There are two other structures in the platypus which also show a long rostro-caudal extension. The striatum extends over 60.2% of the hemisphere length in the platypus and 33.3% in the echidna. This does not agree with the volumetric

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TABLE 3

ABSOLUTE VOLUMES OF VARIOUS BRAIN STRUCTURES AND THEIR PERCENTAGE OF TOTAL BRAIN VOLUME

	PLATYPUS		ECHIDNA		OPOSSUM	
	Abs. vol. mm ³	% of total brain	Abs. vol. mm ³	% of total brain	Abs. vol. mm ³	% of total brain
Neo	4099.28	47.83	11,372.4	42.79	1442.88	22.24
Scz	121.34	1.41	396.9	1.49	—	—
Hip	506.16	5.90	1773.5	6.65	527.88	8.14
Sep	52.33	0.61	108.6	0.41	59.82	0.92
Pal	505.56	5.89	5374.5	20.33	779.75	12.02
Bul	71.37	0.83	835.8	3.14	572.13	8.82
Str	478.21	5.58	1553.5	6.68	240.30	3.70
Di	1050.97	12.27	1768.3	5.84	607.82	9.37
Mes	163.56	1.91	686.7	2.58	373.03	5.75
Cbl	919.55	10.74	1589.3	5.98	839.07	12.93
Med	163.56	1.91	686.7	2.58	572.13	8.82
Mes	—	—	—	—	473.08	7.29
	8571.45	100.00	26,578.2	100.00	6487.89	100.00

The structures in order from top of column are: neocortex, schizocortex, hippocampus, septum, paleocortex, bulbus olfactorius, striatum, diencephalon, mesencephalon, cerebellum, medulla oblongata and mesocortex.

measurements in Table 3. The hippocampus has a rostrocaudal extension of 54.2% of the hemisphere in the platypus and 42.6% in the echidna. This again differs from the volumetric measurements. The misfits between lengths and volumes are of course due to differences in the shape of the structures.

VOLUMETRIC MEASUREMENTS

Table 3 sets out the absolute volumes of each of the eleven brain structures measured and the percentage that that volume represents of the total brain volume. In previous studies on placental animals a placental with little neocortex was used as a "basal" animal with which to compare the degree of development of brain structures in the "higher" placentals, (see Stephan and Pirlot 1970 for bats, Stephan and Andy 1964 for primates). We have no 'basal' monotreme. However, for interest, Table 4 shows the relative percentage of the various brain structures for 'basal' insectivores, the platypus, the echidna, the American opossum and the prosimians (basal insectivore and prosimian data from Stephan and Andy 1964).

1. The neocortex of the platypus is slightly larger relatively than the echidna, and this is surprising as the platypus brain is lissencephalic while that of the echidna is gyrencephalic.

2. The striatum and the hippocampus are of equal size relatively though slightly larger in the echidna in spite of the greater rostro-caudal extension of both components in the platypus.
3. The bulbus olfactorius is relatively underdeveloped in the platypus compared to the echidna, which might be expected for an aquatic animal.
4. Similarly the paleocortex (+ amygdala) is smaller, relatively, in the platypus than in the echidna.
5. The diencephalon is much larger in the platypus, probably as a result of its greater sagittal extension in the platypus.
6. The mesencephalon is relatively small in both animals when compared with many other mammals.
7. The larger relative size of the cerebellum in the platypus is no doubt related to its three dimensional movements in water.

TABLE 4

VOLUMES OF SUBDIVISIONS OF THE BRAIN IN % TOTAL BRAIN

	BASAL INSECT	PLATYPUS	ECHIDNA	PROSIMIAN
NEO	13	48	43	54
SCZ	3	1	1	2
HIP	9	6	7	4
SEP	2	1	1	1
PAL	19	6	20	4
BUL	10	1	4	2
STR	5	5	6	5
DI	8	12	6	8
MES	6	2	2	3
CBL	12	11	6	12
MED	13	7	4	5

Abbreviations as in Table 3.

DISCUSSION

1. DIFFERENTIATION OF THE NEOCORTEX

The various layers in the neocortex of the platypus and the echidna are roughly similar although some differences in thickness and density do exist. The most important difference is probably the greater abundance of large (sometimes pyramidal) neurons in the medial half of layer V in the echidna. We are inclined to believe that this suggests an advance in cortical organisation. This need not necessarily also suggest that the echidna is a more progressive mammal than the platypus.

2. PRESENCE OF A SCHIZOCORTEX

The restriction, in both monotremes, of the schizocortex to the subicular-retrosplenial areas is remarkable. By the way of contrast, the bats have a schizocortex that may extend behind and around the hippocampal formation to the lateral edge of the brain, between paleo- and neo-cortex. It is generally agreed that all schizocortex is of archipallial origin. The medial (subicular) portion represents an earlier development than the lateral one, probably both in ontogeny and in phylogeny. This is not surprising considering the original dorso-central and paramedial position of the archipallium. The extension of that component, or of its derivative (schizocortex) towards the lateral regions, probably is the final stage of its quantitative and topographic evolution before it became reduced in size in higher mammals — (see Stephan and Andy 1970).

3. IDENTIFICATION OF AN INTERMEDIATE CORTEX

The most striking cortical difference between our monotreme brains and that of *Didelphis* consists in the latter possessing a very distinct *piriform cortex* between paleo- and neo-cortex (see Oswaldo-Cruz and Rocha-Miranda 1968). As mentioned previously, the piriform region is characterised by a double layer II of very densely packed cells. Such a high accumulation of cells, most of them oriented in the same direction, reminds one of an immature, even embryonic, tissue. Pirlot (1975), in a study of the South American Edentate *Dasypus* tentatively termed such an organisation "mesocortex". We did not find a distinct piriform mesocortex in the monotremes studied, but the organisation of layer II in both monotremes reminds one of an "unachieved cortex" — a condition close to that of a "mesocortex". There is a common relatively primitive appearance of piriform, mesocortical and monotreme cortices. Thus, although the cortices of lower placentals, marsupials and monotremes each show unique characteristics, they share the same general form. In general, the platypus and the echidna have a relatively undifferentiated cortex and may be considered "paramesocortical" animals.

4. OLFACTION

A special remark must be made concerning the olfactory structures of the brains studied in this paper. Considering firstly the bulbs, from *Platypus* (0.83%) to *Tachyglossus* (3.14%) and *Didelphis* (8.82%), we have an ascending order of size that probably corresponds to an increasing order of use of that organ. A similar hierarchy (with more modest figures ranging from 1.61 to 3.92%) was found previously among 43 species of bats showing various degrees of usage of olfaction (Pirlot and Pottier, 1977). In Edentates too, the same observation was made in the case of a species using (6.7%) and one not using (2.3%) olfaction for food detection (Pirlot 1975). But it is important to note that, in all mammals mentioned above, there is no necessary close correlation between the size of the so-called rhinencephalon (paleocortex + amygdaloid region), the size of the bulbus olfactorius and the observed olfactory ability of the animal. We would say that an animal with good olfactory capability has above all a well developed peri-

pheral organ, i.e. olfactory bulbs. Behavioural reduction of olfaction goes along with reduction in bulb size. It is likely to be correlated also with some limited reduction of the rhinencephalon. However, the latter may be quite large even in the case of moderate use of olfaction. Its oversize in *Tachyglossus* is nevertheless intriguing.

5. PRIMITIVENESS AND PROGRESSIVENESS

It should be emphasised that it is very difficult to isolate criteria that clearly establish the "primitiveness" of monotreme brains. However, it is important to attempt to determine which criteria are pleisomorphic and which are apomorphic. In placentals, progressiveness, is measured by the progression index which is the ratio of brain size in the animal examined compared to the brain size of a 'basal insectivore' of the same body weight. We have no reference type for an allometric comparison of brain components in Monotremes.

The cortex could be considered to be among the most primitive mammalian cortices on the basis of the low number and low density of large, especially pyramidal, neurons. It is thus surprising to find that a very high proportion of cortex is neocortex. This does not necessarily mean an advanced degree of progressiveness, although the two are usually related. Among the chiroptera, Stephan and Pirlot (1970) and Pirlot and Pottier (1977) found the following serial values for the neocortical percentages and progression indices: insect eaters 18.85% and 202; nectar feeders 22.84% and 326; frugivorous microchiroptera, 26.38% and 376; frugivorous megachiroptera 29.34% and 429; piscivorous 31.32% and 549; carnivorous 29.95% and 630; sanguivorous 34.50% and 667. The neocortical percentage for the platypus is 47.83% and for the echidna 42.79%, so that we would expect that the progression indices for these two species, if such indices were available, would be about equal. Jerison (1973) uses another ratio, encephalisation quotient in comparing the relative volumes of mammalian species. His figures are; basal insectivores 0.25-0.40; monotremes 0.50-0.75; prosimians 0.7-2.0; higher primates 1.5-7.8.

In all methods, there is a clear indication that monotremes have relatively high brain and neocortical indices. Size alone cannot be taken as a criterion for progressiveness unless a group of related forms, that is forms having a broadly similar organisational brain structure, are being considered. There are obvious qualitative differences between the brains of monotremes and other mammals and this of course will prevent simple comparisons between the two on the basis of quantitative component analysis. It is only within the general framework of a common structural type that quantitative differences can be related to ecology and behaviour, as was done in the studies on chiroptera. Further as Pirlot (1970) showed in a review of data from Insectores, Primates and Chiroptera, even the brain to body ratio cannot be interpreted in a purely quantitative way without considering the differing life habits of the animals.

Thus the two species both show a high neocortex percentage, with the platypus only slightly higher. Most of the brain structures are equally developed in both

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forms, except that the echidna has greater development of the smell areas (bulbus olfactorius and paleocortex) while the platypus has greater development of the cerebellum. This is what one would expect, smell being important to the terrestrial echidna but relatively unimportant to the water and burrow living platypus, whereas a sense of balance is of greater importance to a platypus swimming in the water than to the echidna walking on land.

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